



Applicative report: Economic Resilience in OECD Countries

1. Project Objective

The purpose of this project is to analyze economic resilience as a **structural property of economic systems**, rather than as a simple measure of economic growth or short-term performance.

Instead of building a classical ranking or a composite index, the project constructs a **Partial Order (POSet)** among OECD countries using a selected set of structural economic variables. The objective is to identify which economies exhibit structurally stronger configurations under multidimensional criteria related to resilience.

The resulting order is then validated against major systemic shocks such as:

- the 2008 financial crisis,
- the COVID-19 crisis,
- and the recent inflation/energy shock.

The central hypothesis is:

countries that are structurally dominant in the POSet should exhibit greater stability and adaptive capacity during systemic crises.

2. Data Collection and Dataset Structure

All data are collected from the OECD Data Explorer and merged into a single country-year dataset before any separation between ordering and validation variables.

The initial dataset includes all variables considered potentially relevant for:

- structural resilience,
- macroeconomic performance,
- and shock response analysis.

After cleaning and preprocessing, the dataset is divided into two logical subsets:

A. Ordering Variables (used for the POSet)

These variables define the structural economic space used to compare countries.

Variable	Interpretation	Desired Direction
R&D expenditure (% GDP)	Innovation and adaptive capability	Higher is better

Variable	Interpretation	Desired Direction
Trust in institutions	Institutional coordination and social stability	Higher is better
Public debt (% GDP)	Financial fragility and fiscal rigidity	Lower is better
Energy import dependency	External structural vulnerability	Lower is better
Tertiary education attainment	Human capital and workforce adaptability	Higher is better

These variables are intentionally limited in number because POSet structures become unstable and excessively sparse when too many dimensions are included simultaneously.

The selected variables represent:

- relatively slow-moving structural characteristics,
- low redundancy,
- and distinct economic mechanisms.

B. Validation Variables (used after the POSet construction)

These variables are NOT used to construct the order.

They are only used to evaluate whether structurally dominant economies actually behave more resiliently during crises.

Variable	Validation Purpose
GDP growth / contraction	Shock severity
Inflation rate	Macroeconomic instability
Unemployment variation	Labour market deterioration
Productivity variation	Economic continuity
Recovery time	Adaptive speed
Volatility measures	Systemic instability

This separation is fundamental:

- the order must describe structure,
- while validation measures outcomes during shocks.

3. Data Preprocessing

Before constructing the POSet, the entire dataset undergoes preprocessing.

The preprocessing pipeline includes:

- missing value handling,
- country-year alignment,
- duplicate removal,
- directional consistency,

- normalization.

Directional consistency means that all variables are transformed so that:

- larger values always correspond to structurally preferable conditions.

For example:

- debt and energy dependency are inverted before normalization.

Normalization is necessary because variables have different scales and distributions. Robust scaling methods based on median and interquartile range are preferred over standard z-score normalization due to the presence of outliers in macroeconomic data.

4. Construction of the POSet

4.1 Structural Representation

Each country is represented as a multidimensional vector:

$$x = (x_1, x_2, x_3, x_4, x_5)$$

where each component corresponds to one structural ordering variable.

The POSet is therefore a comparison structure built directly in this multidimensional space.

4.2 Dominance Relation

The project uses a robust version of Pareto dominance called **ϵ -dominance**.

A country A dominates country B if:

- A is not worse than B across all structural variables within a tolerance threshold,
- and A is strictly better in at least one variable beyond that threshold.

Formally: $x_i(A) \geq x_i(B) - \varepsilon_i \quad \forall i$

and $x_j(A) > x_j(B) + \varepsilon_j$

for at least one dimension (j).

This tolerance mechanism prevents very small statistical differences from generating artificial dominance relations.

4.3 Definition of ϵ Thresholds

For each variable:

$$\varepsilon_i = 0.1 \cdot \sigma_i$$

where:

- (σ_i) is the empirical standard deviation of variable (i).

This produces:

- more stable comparisons,
- fewer noisy relations,
- and a more interpretable order structure.

4.4 Dominance and Comparability Matrices

Once dominance conditions are computed for every country pair, two matrices are generated.

Dominance Matrix

$$D_{ij} = 1$$

if country (i) dominates country (j).

Comparability Matrix

Countries are comparable if:

- one dominates the other.

Otherwise they are considered incomparable.

Incomparability is not an error: it represents economies with structurally different trade-offs.

4.5 Hasse Diagram and Pareto Frontier

The final POSet is visualized through:

- a Hasse diagram,
- and the Pareto frontier.

The Hasse diagram removes transitive relations and keeps only direct dominance relations, producing a readable structural hierarchy.

The Pareto frontier identifies:

- countries that are not dominated by any other country.

These countries represent structurally strong configurations under the selected dimensions.

Figure on the right is an example of the result.

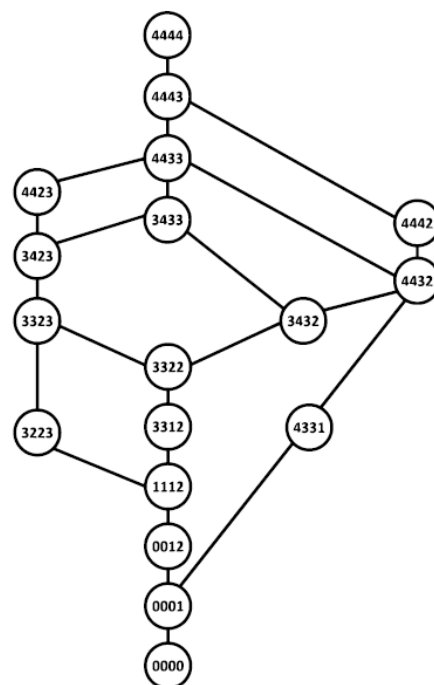


Fig. 1. Hasse diagram of π_{x_i} , the component-wise poset of the unique profiles derived from the PP&P MIS. The diagram must be read from top to bottom: if node y is linked by a downward path of edges to node x , then the two nodes are comparable and $x \leq_x y$; if such a path does not exist, the two nodes are incomparable.

5. Shock-Based Validation

The POSet itself is not built using crisis variables. Instead, it is validated against historical shocks.

The procedure is:

Step 1 — Build a pre-shock POSet

Examples:

- build a 2007 structural POSet,
- build a 2019 structural POSet.

Step 2 — Observe crisis dynamics

Measure:

- GDP contraction,
 - unemployment increase,
 - inflation instability,
 - recovery time,
- during the subsequent crisis period.

Step 3 — Compare structural position and shock behavior

The analysis verifies whether:

- Pareto-dominant countries,
 - or structurally higher countries in the POSet,
- show greater resilience during shocks.
-

6. Expected Contribution

The project contributes a framework that:

- avoids arbitrary weighted indexes,
- avoids forced linear rankings,
- explicitly handles incomparability,
- separates structural ordering from shock outcomes,
- and models resilience as a multidimensional structural phenomenon.

Rather than asking:

| “Which country is the best?”

the project asks:

| “Which structural economic configurations appear more robust under systemic stress?”